Probability & Statistics for Machine Learning & Data Science

Illustrated study notes · Weeks 1 – 4

About these notes. They follow the public syllabus of the DeepLearning.AI course *Probability & Statistics for Machine Learning & Data Science* (course 3 of the Mathematics for Machine Learning and Data Science Specialization). The explanations, examples and diagrams are written independently to make each idea easy to picture. They are not the official course material, so use them next to the lectures and check anything important against them.

How to read the boxes

KEY IDEA
The idea in one or two sentences. If you remember only one thing, remember this.

EXAMPLE
A small worked example with numbers you can redo by hand.

IN YOUR LOAN-DEFAULT PROJECT
Where the idea showed up in the loan-default EDA (default rate 8.07% on 307,511 applicants).

WATCH OUT
A common mistake or misreading.

The course in one picture

Week 1 Probability and distributions	**Week 2** Describing distributions; many variables	**Week 3** Sampling and estimation	**Week 4** Confidence intervals and tests
What is the chance of things? Bayes. Binomial, normal.	Mean, variance, skew. Joint, covariance, multivariate normal.	Samples, CLT, maximum likelihood, MAP.	How sure are we? Is the difference real? A/B tests.

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The thread running through all four weeks: describe uncertainty (weeks 1 – 2), **learn from a sample** (week 3), and **say how confident you are** (week 4).

Week 1 · Probability and probability distributions

1.1 Probability basics

The **probability** of an event is a number between 0 (impossible) and 1 (certain). When all outcomes are equally likely, it is *favourable outcomes ÷ total outcomes*. The set of all possible outcomes is the **sample space**.

> **EXAMPLE**
> **Two dice.** There are 6 × 6 = 36 equally likely outcomes. The sum 7 can happen in 6 ways ((1,6), (2,5), (3,4), (4,3), (5,2), (6,1)), so P(sum = 7) = 6/36 = 1/6.

Complement: “not A”

$$P(\text{not A}) = 1 - P(A)$$

Use it for “at least one” questions: work out the chance that *nothing* happens, then subtract from 1.

> **EXAMPLE**
> **At least one six in four rolls.** The chance of no six in one roll is 5/6, so in four independent rolls it is $(5/6)^4 \approx 0.482$. Therefore P(at least one six) = 1 − 0.482 = **0.518**.

Sum rule: “A or B”

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

If A and B **cannot happen together** (disjoint), the last term is 0 and you simply add. If they overlap, adding counts the overlap twice, so you subtract it once.

> **EXAMPLE**
> Draw one card. P(heart or king) = 13/52 + 4/52 − 1/52 = 16/52. The king of hearts is in both groups, so it is subtracted once.

Independence: “A and B”

$$P(A \text{ and } B) = P(A)\,P(B)$$

Two events are **independent** if knowing one happened tells you nothing about the other. For independent events the probabilities multiply.

> **EXAMPLE**
> Two coin flips: P(two heads) = ½ × ½ = ¼. **Dependent case:** drawing two cards without putting the first back. P(two aces) = 4/52 × 3/51 — the second probability changed because the deck changed.

The birthday problem

How many people do you need before it is more likely than not that two share a birthday? Most people guess about 180. The answer is **23**. Use the complement: compute the chance that *all* birthdays are different, then subtract from 1.

$$P(\text{shared}) = 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^{n}}$$

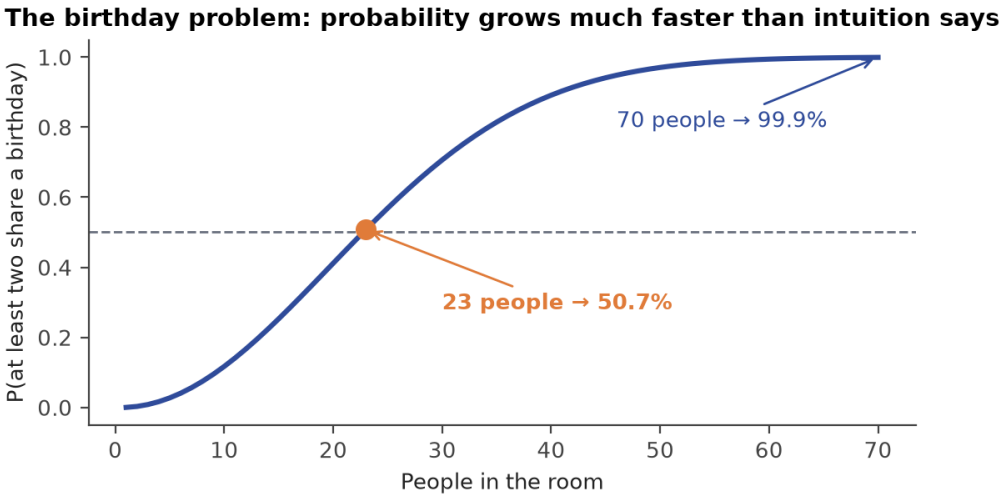


Figure 1.1 · With 23 people the chance already passes 50%. There are many pairs (253 of them), not just 23.

KEY IDEA

Rare-looking coincidences are common once you count how many chances there are. The same logic explains why testing hundreds of columns will throw up some “significant” results by luck (Week 4).

1.2 Conditional probability

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

Read it as “the probability of A *once we know* B happened”. Knowing B shrinks the world to just the B outcomes, and we ask what share of that smaller world is also A.

EXAMPLE

One die. P(six | the roll is even). The even outcomes are {2, 4, 6}, and one of those three is a six. So P(6 | even) = 1/3, while P(6) on its own is 1/6.

A and B are independent exactly when **P(A | B) = P(A)**: learning B does not change your belief about A.

IN YOUR LOAN-DEFAULT PROJECT

The default rate among applicants with a missing EXT_SOURCE_3 is a conditional probability, P(default | missing) ≈ 9.3%, against P(default | present) ≈ 7.8%. The “gap” you measured is a difference of two conditional probabilities. If missingness were independent of default, the gap would be zero.

1.3 Bayes' theorem

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

Bayes' theorem turns a probability you **know** (how likely the evidence is if A is true) into one you **want** (how likely A is, given the evidence).

Term	Name	Meaning in the spam example
P(A)	**Prior**	Before seeing the email: 30% of mail is spam.
P(B | A)	**Likelihood**	If it is spam, the word “free” appears 60% of the time.
P(B)	**Evidence**	How often “free” appears in any email (spam or not).
P(A | B)	**Posterior**	After seeing “free”, how likely is spam?

The evidence term is usually built from the **total probability** rule:

$$P(B) = P(B \mid A)\,P(A) + P(B \mid \text{not } A)\,P(\text{not } A)$$

EXAMPLE

Spam filter. 30% of emails are spam. “free” appears in 60% of spam and 5% of ordinary mail.
P(free) = 0.60 × 0.30 + 0.05 × 0.70 = 0.18 + 0.035 = 0.215
P(spam | free) = 0.18 / 0.215 = **0.837**.

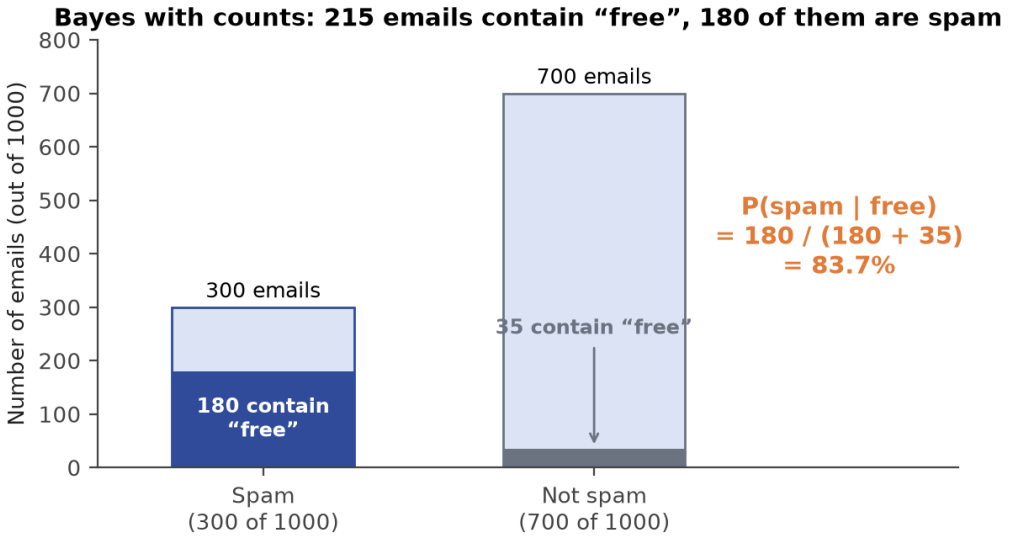


Figure 1.2 · The same calculation with counts out of 1000 emails. Of the 215 emails containing “free”, 180 are spam.

WATCH OUT

Base rates matter. A very accurate test for a rare condition still gives mostly false alarms, because the prior P(A) is tiny. Always ask “how common is this to begin with?”.

Naive Bayes classifier

With many features, estimating P(features | class) directly is impractical. **Naive Bayes** assumes the features are independent *given the class*, so the likelihoods simply multiply:

$$P(\text{class} \mid x_1, \ldots, x_n) \;\propto\; P(\text{class}) \prod_{i=1}^{n} P(x_i \mid \text{class})$$

The assumption is “naive” because real features are rarely independent, yet the classifier often works well and is very fast. You pick the class with the largest value. This is the Week 1 programming assignment (a spam classifier).

KEY IDEA

Probability in machine learning. A classifier does not just say “default” or “no default”. Its natural output is P(default | applicant's features), a probability you can threshold, rank and calibrate.

1.4 Random variables and distributions

A **random variable** turns the outcome of a random process into a number, for example “the number of defaults among 100 loans”. A **distribution** says how probability is spread over its possible values.

Discrete distributions

Countable values, each with a probability; the probabilities add to 1.

> **KEY IDEA**
> **Bernoulli(p):** one trial, 1 with probability p, 0 otherwise.
> **Binomial(n, p):** the number of successes in n independent Bernoulli(p) trials. Bernoulli is the special case n = 1.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

The coefficient C(n, k) = n! / (k!(n − k)!) counts the ways to place k successes among n trials. Mean and variance:

$$E[X] = np, \qquad \mathrm{Var}(X) = np(1-p)$$

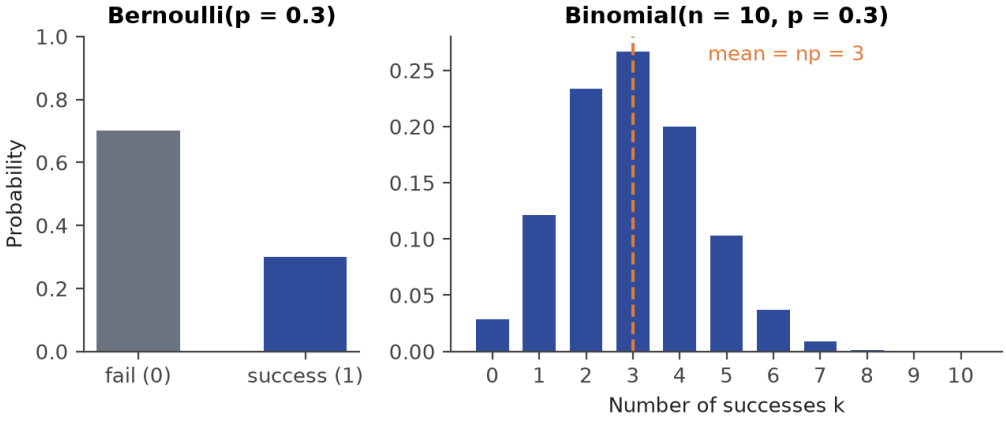


Figure 1.3 · Left: a single Bernoulli trial. Right: the number of successes in ten trials, centred on np = 3.

> **IN YOUR LOAN-DEFAULT PROJECT**
> Whether a given applicant defaults is Bernoulli with p ≈ 0.081. The number of defaults among n applicants is Binomial(n, 0.081): for n = 1,000 you expect about 81, give or take roughly 8.6 (√(np(1 − p))).

Continuous distributions

Values on a continuum (income, time, a score between 0 and 1). The probability of landing on one exact value is 0, so we talk about **ranges**.

- **PDF** f(x), the density: P(a ≤ X ≤ b) is the **area under the curve** between a and b. The total area is 1.
- **CDF** F(x) = P(X ≤ x): the area to the left of x. It rises from 0 to 1, and P(a ≤ X ≤ b) = F(b) − F(a).

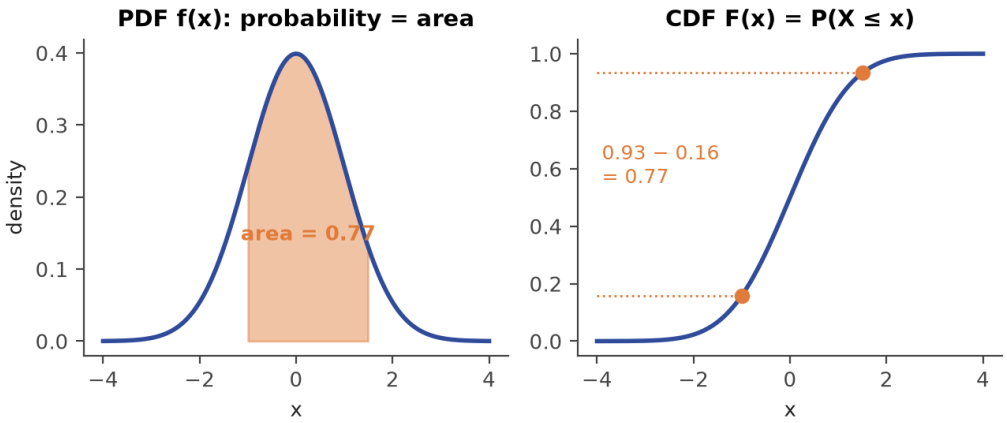


Figure 1.4 · The shaded area in the PDF (left) equals the difference of two CDF heights (right).

Distribution	Shape / use	Parameters
Uniform	Flat: every value in an interval is equally likely (random numbers between 0 and 1).	lower and upper limit
Normal (Gaussian)	Symmetric bell curve. Appears when many small effects add up (heights, measurement error, averages).	mean μ, standard deviation σ
Chi-squared (optional)	Sum of squared standard normals. Skewed, only positive. Behind chi-square tests and variance estimates.	degrees of freedom

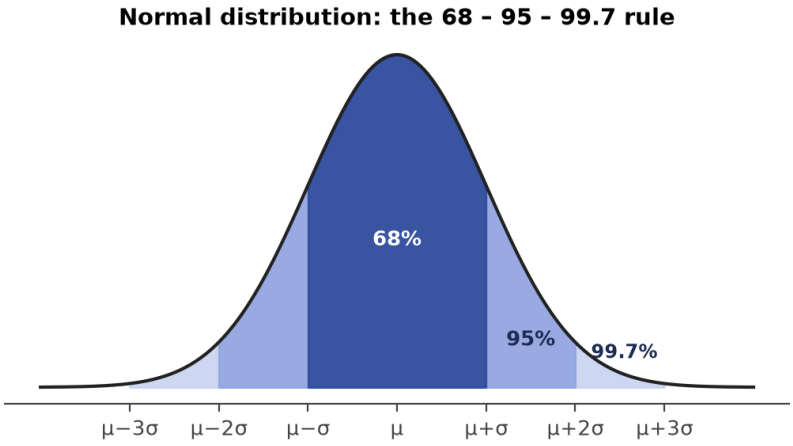


Figure 1.5 · The 68 – 95 – 99.7 rule: share of a normal distribution within 1, 2 and 3 standard deviations of the mean.

> **EXAMPLE**
> **Heights** with mean 170 cm and σ = 10 cm. About 95% of people fall between 150 and 190 cm (μ ± 2σ), and only about 0.13% are taller than 200 cm (3σ above the mean).

Sampling from a distribution means drawing random values that follow it, for example with *numpy.random.normal*. Simulation is the quickest way to build intuition: draw a thousand values and plot a histogram.

Check yourself (answers in Appendix C)

- Why is “at least one six in four rolls” easier through the complement? What is the answer?
- If A and B are independent, what is P(A | B)?
- In the spam example, which number is the prior, the likelihood and the posterior?
- A continuous variable has P(X = 5) = 0, yet P(4 ≤ X ≤ 6) can be large. Why?
- For Binomial(n = 10, p = 0.3), what are the mean and the variance?

Week 2 · Describing distributions and multiple variables

2.1 Expected value

The **expected value** E[X] is the long-run average of a random variable: each value weighted by its probability.

$$E[X] = \sum_x x\, p(x) \qquad E[X] = \int x\, f(x)\, dx$$

> **EXAMPLE**
> **A fair die:** E[X] = (1 + 2 + 3 + 4 + 5 + 6) × 1/6 = 3.5. You never roll a 3.5, but it is the average over many rolls.

Mean, median, mode. The mean is the balance point. The **median** is the middle value (half the probability on each side). The **mode** is the most likely value. For a skewed distribution the mean is pulled toward the long tail.

Expected value of a function works the same way: apply the function to each value, then weight by probability.

$$E[g(X)] = \sum_x g(x)\, p(x)$$

Linearity is the rule you will use constantly. It holds even when the variables are dependent:

$$E[aX + b] = aE[X] + b, \qquad E[X + Y] = E[X] + E[Y]$$

> **EXAMPLE**
> **Expected profit on a loan** (illustrative numbers). The borrower repays with probability 0.92, earning 800 in interest, and defaults with probability 0.08, losing 10,000.
> E[profit] = 0.92 × 800 + 0.08 × (−10,000) = 736 − 800 = **−64**.
> On average this loan loses money, which is why predicting default well matters.

2.2 Variance and standard deviation

Variance measures spread: the average squared distance from the mean. The **standard deviation** σ = √Var is in the original units, so it is easier to read.

$$\mathrm{Var}(X) = E\left[(X - \mu)^2\right] = E[X^2] - \mu^2$$

> **EXAMPLE**
> **A fair die:** $E[X^2]$ = (1 + 4 + 9 + 16 + 25 + 36)/6 = 91/6 ≈ 15.17, so Var = 15.17 − 3.5^2 = 35/12 ≈ **2.92** and σ ≈ 1.71.

Useful rules

$$\mathrm{Var}(aX + b) = a^2\, \mathrm{Var}(X)$$

- Adding a constant moves the distribution but does not change its spread; scaling by a stretches the spread by |a|.

- For **independent** variables, Var(X + Y) = Var(X) + Var(Y). Variances add; standard deviations do not.
- Bernoulli(p): mean p, variance p(1 − p). Binomial(n, p): mean np, variance np(1 − p).

Sum of Gaussians. The sum of independent normal variables is again normal, and means and variances add:

$$X + Y \sim N(\mu_X + \mu_Y,\ \sigma_X^2 + \sigma_Y^2) \quad \text{(independent)}$$

Standardizing: the z-score

$$z = \frac{x - \mu}{\sigma}$$

Subtracting the mean and dividing by σ turns any variable into one with mean 0 and standard deviation 1. A z-score says “how many standard deviations from average”. It puts features on a common scale.

EXAMPLE
Heights have μ = 170 cm and σ = 10 cm. Someone 190 cm tall has z = (190 − 170)/10 = **2**, two standard deviations above average.

IN YOUR LOAN-DEFAULT PROJECT
Scaling features (standardizing) matters for linear models and neural networks, because a column in the hundreds of thousands (income) would otherwise dominate one between 0 and 1 (EXT_SOURCE scores).

2.3 Skewness and kurtosis

The mean and variance do not capture the shape. Two more summaries do:

- **Skewness** = $E[((X - \mu)/\sigma)^3]$ measures asymmetry. **Positive** skew means a long right tail (mean > median). **Negative** skew means a long left tail.
- **Kurtosis** = $E[((X - \mu)/\sigma)^4]$ measures how heavy the tails are. A normal distribution has kurtosis 3, so people often report *excess kurtosis* = kurtosis − 3. Higher values mean more extreme outliers.

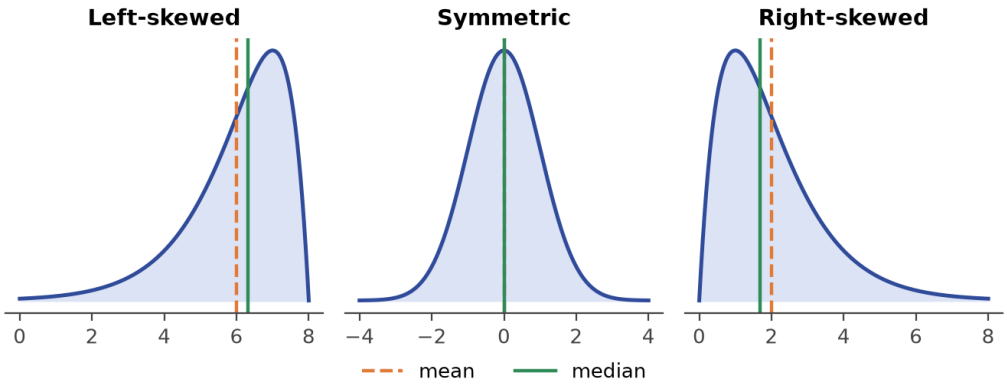


Figure 2.1 · In a right-skewed distribution the mean (dashed) is pulled out to the tail, beyond the median.

IN YOUR LOAN-DEFAULT PROJECT
AMT_INCOME_TOTAL is strongly right-skewed: its mean (about 168,800) is far above a typical value and a single 117,000,000 record sits in the tail. For such columns a **log transform** usually makes the shape closer to symmetric.

2.4 Quantiles and ways to visualise a distribution

The **p-th quantile** is the value below which a share p of the data falls. The 50th percentile is the median; the 25th and 75th are the **quartiles**, and the **interquartile range** IQR = Q3 − Q1 contains the middle half. A common outlier rule flags values beyond Q1 − 1.5 × IQR or Q3 + 1.5 × IQR.

Plot	What it shows	Good for
Box plot	Median, quartiles, whiskers and points beyond them	Quick summary, comparing groups, spotting outliers
Kernel density estimate (KDE)	A smooth histogram	The shape without choosing bin widths
Violin plot	A KDE mirrored around a box plot	Shape and summary together, per group
QQ plot	Sample quantiles against the quantiles of a reference distribution (often normal)	Checking normality: points on the line mean “normal”

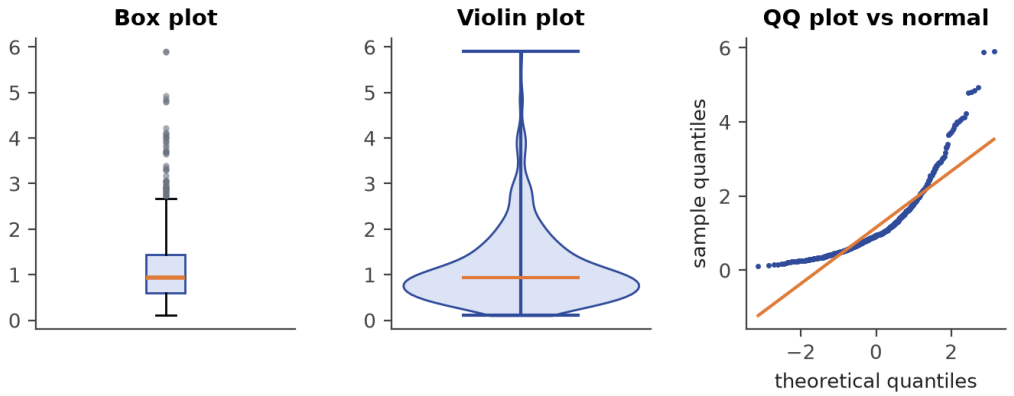


Figure 2.2 · The same right-skewed sample. The curved QQ plot (right) reveals that it is not normal.

WATCH OUT

Many methods assume roughly normal data. Check with a QQ plot or histogram before trusting them, and transform (for example with a log) if the shape is badly skewed.

2.5 Joint, marginal and conditional distributions

With two variables we describe them together. The **joint** distribution gives the probability of each combination. The **marginal** distribution of one variable is found by summing over the other. The **conditional** distribution fixes one variable and asks about the other: **conditional = joint ÷ marginal**.

IN YOUR LOAN-DEFAULT PROJECT

Joint table for EXT_SOURCE_3 (missing or present) and default (rounded, from the EDA numbers):

	Default	No default	Row total (marginal)
EXT_SOURCE_3 missing	0.018	0.180	0.198
EXT_SOURCE_3 present	0.062	0.740	0.802
Column total (marginal)	0.080	0.920	1.000

- **Marginal:** P(missing) = 0.018 + 0.180 = 0.198. About 20% of applicants lack the score.
- **Conditional:** P(default | missing) = 0.0185 / 0.198 ≈ **9.3%**, and P(default | present) = 0.0623 / 0.802 ≈ **7.8%**.
- **Independence test:** if independent, P(default and missing) would equal P(default) × P(missing) = 0.081 × 0.198 ≈ 0.016. The actual 0.0185 is higher, so missingness and default are **dependent**.

For continuous variables the same ideas hold with densities: integrate (instead of sum) to get a marginal.

2.6 Covariance and correlation

Covariance says whether two variables move together: positive if they tend to be above or below their means at the same time, negative if one is high when the other is low.

$$\mathrm{Cov}(X, Y) = \mathrm{E}[(X - \mu_X)(Y - \mu_Y)]$$

Its size depends on the units, so **correlation** divides by both standard deviations. The result is always between −1 and 1:

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \, \sigma_Y}$$

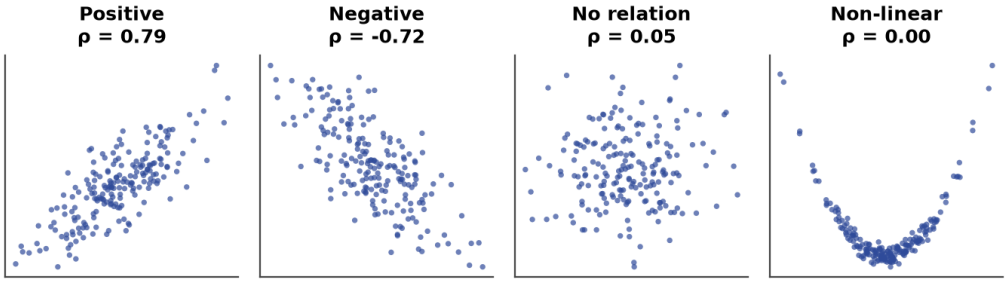


Figure 2.3 · Positive, negative, none, and a strong non-linear relation with ρ ≈ 0.

WATCH OUT

Zero correlation does not mean independent. Correlation only detects *linear* relationships. In the right-hand panel y depends completely on x ($y = x^2$ plus noise) but ρ is about 0. Independent variables do have zero covariance; the reverse is not true. And correlation is not causation.

IN YOUR LOAN-DEFAULT PROJECT

The Spearman correlation between AMT_REQ_CREDIT_BUREAU_MON and EXT_SOURCE_3 was −0.019, essentially zero. Their *values* are unrelated even though their *missingness* overlaps strongly. Always ask which of the two you are measuring.

Covariance matrix. For several variables, collect all variances and covariances in one symmetric matrix Σ:

	X	Y
X	Var(X)	Cov(X, Y)
Y	Cov(X, Y)	Var(Y)

The diagonal holds the variances; the off-diagonal entries are the covariances. It is the starting point for PCA.

2.7 The multivariate Gaussian

The normal distribution generalises to several variables. It is described by a **mean vector** (the centre) and a **covariance matrix** (the shape). Its contours are ellipses: circles when the variables are uncorrelated, tilted ellipses when they are correlated.

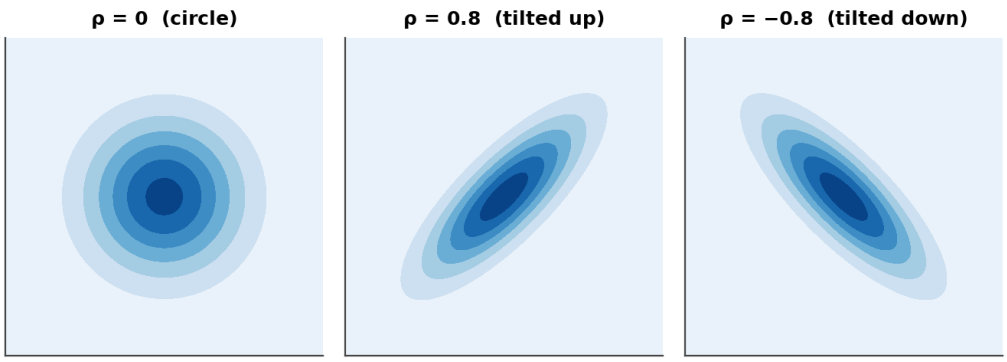


Figure 2.4 · The sign and size of the correlation set the tilt and stretch of the ellipse.

KEY IDEA

Many ML models (Gaussian mixtures, discriminant analysis, Gaussian processes, PCA) are built on this distribution, because it is fully specified by just a mean vector and a covariance matrix.

Check yourself (answers in Appendix C)

- What are the expected value and variance of a fair die?
- In a right-skewed distribution, which is larger, the mean or the median?
- If X and Y are independent, what is Cov(X, Y)? Is the converse true?
- How does correlation differ from covariance?
- From the joint table, what is P(default | EXT_SOURCE_3 missing)?

Week 3 · Sampling and point estimation

3.1 Population, sample and statistics

The **population** is everything you care about; a **sample** is the part you actually observe. A number describing the population is a **parameter** (μ, σ, p). A number computed from the sample is a **statistic**, and it is your **estimate** of the parameter.

Quantity	Formula	Estimates
Sample mean	$\bar{x} = (1/n)\ \Sigma\ x_i$	population mean μ
Sample proportion	$\hat{p}$ = (number of successes) / n	population proportion p
Sample variance	$s^2 = (1/(n-1))\ \Sigma\ (x_i - \bar{x})^2$	population variance σ^2

KEY IDEA

Why n − 1? The deviations are measured from $\bar{x}$, which was itself computed from the same data, so they are a little smaller than deviations from the true mean. Dividing by n − 1 instead of n corrects that bias. (This is called Bessel's correction.)

IN YOUR LOAN-DEFAULT PROJECT

The 8.07% default rate is a sample proportion $\hat{p}$ computed from 307,511 applicants. It estimates the true default probability of the underlying population of applicants.

3.2 Law of large numbers

As the sample grows, the sample mean settles on the true mean. More data means a more reliable estimate.

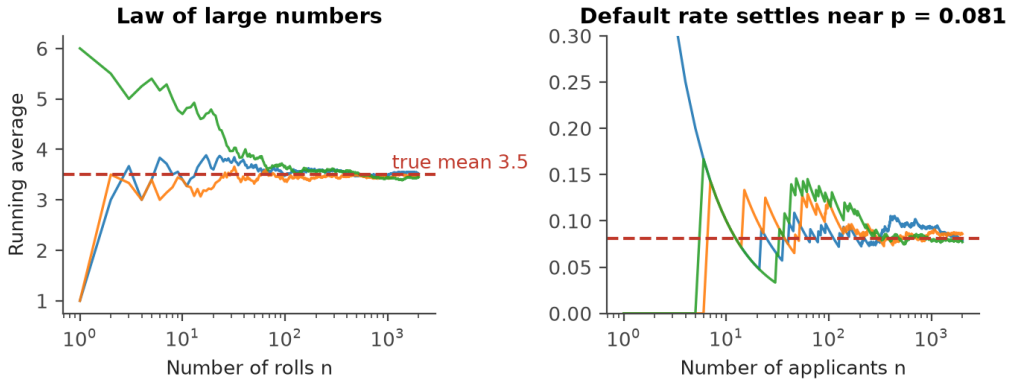


Figure 3.1 · Left: running averages of die rolls approach 3.5. Right: the default rate stabilises near 8.1% as more applicants are included.

WATCH OUT

The law of large numbers is about the average over many trials. It does not say that a streak of bad luck is “due” to reverse (the gambler's fallacy). Each flip is still independent.

3.3 The central limit theorem (CLT)

Take many samples of size n from *any* population (with finite variance) and compute each sample's mean. Those means form a distribution that is approximately **normal**, centred at μ, with a spread that shrinks as n grows:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right), \qquad SE = \frac{\sigma}{\sqrt{n}}$$

The standard deviation of the sample mean is the **standard error** SE = σ/√n.

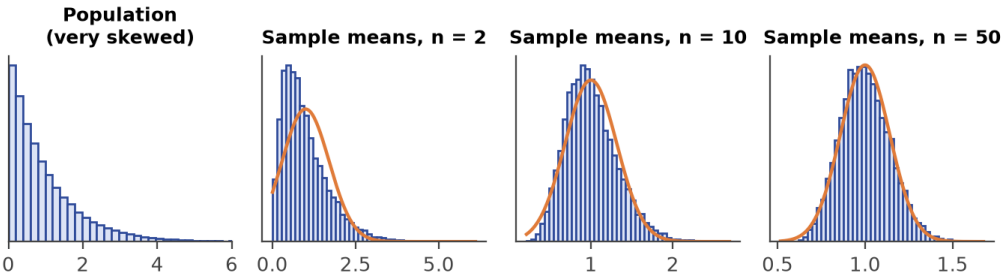


Figure 3.2 · The population is very skewed, yet the distribution of sample means becomes bell-shaped as n grows (orange: the normal curve the CLT predicts).

KEY IDEA

Why it matters. It lets us use the normal distribution to reason about averages and proportions even when the raw data are not normal. A common rule of thumb is n ≥ 30, though very skewed data need more.

EXAMPLE

Quadrupling the sample size halves the standard error, because of the √n. To be twice as precise you need four times the data.

IN YOUR LOAN-DEFAULT PROJECT

For a proportion p̂ the CLT gives SE = √(p(1 − p)/n). That is the formula behind the “se_pts” column in your EDA tables, and the reason “gap > 2 × SE” was used as a rough test of whether a gap is real.

3.4 Point estimation and maximum likelihood (MLE)

A **point estimate** is a single best guess for a parameter. **Maximum likelihood** picks the parameter value that makes the observed data *most probable*.

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$$

Because products of many small numbers are awkward, we maximise the **log-likelihood** instead. It has its peak at the same place:

$$\ell(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

EXAMPLE

A coin shows 7 heads in 10 flips. The likelihood of heads-probability p is $p^7(1-p)^3$. It peaks at **p = 0.7**, which is simply the observed share, k/n.

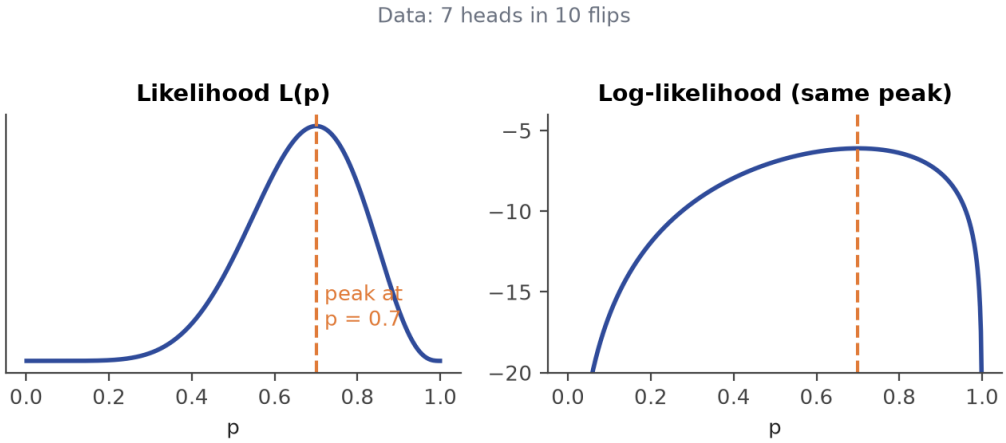


Figure 3.3 · The likelihood (left) and its logarithm (right) peak at the same value, p = 0.7.

Model	Maximum-likelihood estimate
Bernoulli / binomial	$\hat{p} = k / n$ (the sample proportion)
Gaussian	$\hat{\mu} = \bar{x}$ (the sample mean) and $\hat{\sigma}^2 = (1/n)\ \Sigma\ (x_i - \bar{x})^2$ (divides by n, so it is slightly biased)
Linear regression with Gaussian noise	Maximising the likelihood is the same as **minimising the squared error** (least squares)

$$\hat{p}_{\text{MLE}} = \frac{k}{n}$$

KEY IDEA

The squared-error loss you use to train a linear regression is not arbitrary. It is the maximum-likelihood fit when the noise is assumed to be Gaussian. Many ML loss functions are negative log-likelihoods in disguise.

3.5 Regularization

A model fitted purely by maximum likelihood can overfit: it chases noise in a small sample with large weights. **Regularization** adds a penalty on the size of the weights to the loss, trading a little fit for stability:

$$\min_{w} \sum_{i} (y_i - w^T x_i)^2 \ + \ \lambda \|w\|^2$$

The knob λ controls the trade-off: λ = 0 is plain MLE, a large λ shrinks the weights strongly toward zero. This form is called **L2 (ridge)** regularization; penalising $\Sigma|w_i|$ instead gives **L1 (lasso)**.

3.6 Bayesian statistics

	Frequentist	Bayesian
The parameter is...	a fixed, unknown number	uncertain, described by a distribution
Probability means...	long-run frequency over repeated samples	degree of belief, updated by data
Output	a point estimate (and a confidence interval)	a full posterior distribution

Bayesian inference is Bayes' theorem applied to a parameter θ: start with a **prior** belief, multiply by the **likelihood** of the data, and you get the **posterior**:

$$p(\theta \mid x) \propto p(x \mid \theta)\, p(\theta)$$

MAP (maximum a posteriori) estimation picks the *peak of the posterior*:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \left[\log p(x \mid \theta) + \log p(\theta)\right]$$

EXAMPLE

Coin, continued. Before flipping, you believe the coin is probably fair: a Beta(2, 2) prior. After 7 heads and 3 tails the posterior is Beta(2 + 7, 2 + 3) = Beta(9, 5). The MLE is 7/10 = 0.70, but the MAP is $(9 - 1)/(9 + 5 - 2) = 8/12 \approx$ **0.667**: the prior pulls the estimate a little toward 0.5.

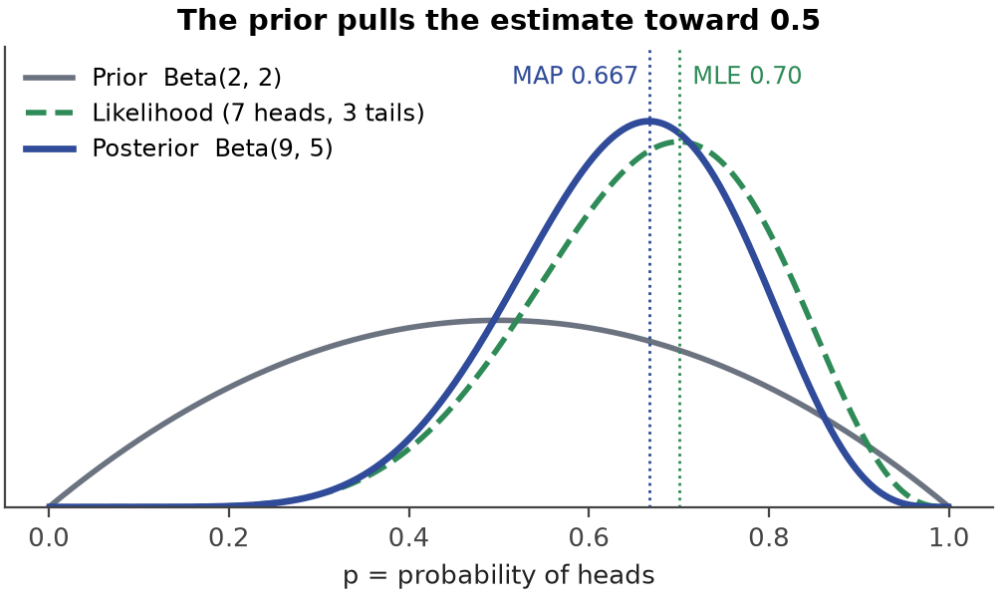


Figure 3.4 · The posterior (blue) sits between the prior (grey) and the likelihood (green). With more data the likelihood wins.

How MAP, MLE and regularization connect

Prior on the weights	MAP estimate equals…
Flat (uniform) prior: no preference	the MLE
Gaussian prior centred at 0	MLE + **L2** (ridge) penalty
Laplace prior centred at 0 (beyond the course)	MLE + **L1** (lasso) penalty

KEY IDEA

Regularization is the Bayesian idea in disguise: it says “before seeing the data, I believe the weights are small”. As the sample grows, the data outweigh the prior and MAP approaches MLE.

Check yourself (answers in Appendix C)

- Why does the sample variance divide by n − 1?
- What does the CLT say about the sample mean when n = 50, even for a skewed population?
- What is the MLE of p after 7 heads in 10 flips?
- With a Gaussian prior on the weights, what does the MAP estimate correspond to?
- If you quadruple the sample size, what happens to the standard error?

Week 4 · Confidence intervals and hypothesis testing

4.1 Confidence intervals

A point estimate (a single number) says nothing about how reliable it is. A **confidence interval** gives a range of plausible values for the parameter, built from the estimate and its standard error.

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

z* depends on the confidence level, and the **margin of error** (ME) is the part after the ± sign.

Confidence level	90%	95%	99%
Critical value z*	1.645	1.96	2.576

EXAMPLE

A sample of n = 36 has mean 52 and the known σ = 12. SE = 12/√36 = 2, ME = 1.96 × 2 = 3.92, so the 95% interval is **(48.1, 55.9)**.

What “95% confident” really means

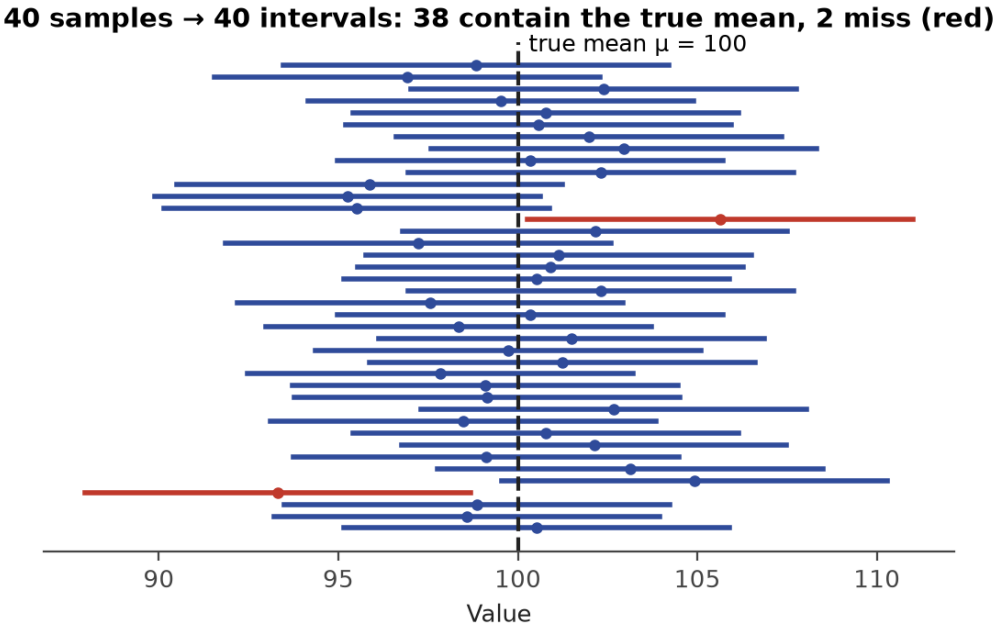


Figure 4.1 · Forty samples give forty intervals. About 95% of them contain the true mean; here 38 do and 2 miss (red).

WATCH OUT

It does **not** mean “there is a 95% probability that the true value lies in *this* interval”. The true value is fixed. The 95% describes the *procedure*: if you repeated the sampling many times, 95% of the intervals would capture it.

Choosing the sample size

$$n = \left(\frac{z^* \sigma}{ME}\right)^2$$

EXAMPLE
To estimate a mean to within ±2 when σ = 15 at 95% confidence: $n = (1.96 \times 15 / 2)^2 = 14.7^2 \approx 216.1$, so you need **217** observations.

When σ is unknown: the t interval

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

In practice σ is unknown and is replaced by the sample value s. That adds uncertainty, so we use the **t distribution** with n − 1 degrees of freedom (section 4.3), which gives a slightly wider interval.

EXAMPLE
n = 16, $\bar{x} = 52$, s = 12. SE = 12/√16 = 3, t* for 15 degrees of freedom at 95% is 2.131, so ME = 6.39 and the interval is **(45.6, 58.4)**.

Confidence interval for a proportion

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

IN YOUR LOAN-DEFAULT PROJECT
Default rate: $\hat{p} = 0.0807$ with n = 307,511 gives SE = 0.00049 and ME = 0.00096. The 95% interval is **7.97% to 8.17%**, a very tight estimate.
A tiny group: FLAG_DOCUMENT_2 had 13 minority rows with 4 defaults, $\hat{p} = 30.8\%$. The interval is roughly **5.7% to 55.9%**, so wide that the data cannot distinguish it from the 8% base rate (the normal approximation is only rough at n = 13, but the message stands). That is the statistical reason for not trusting columns with a handful of rows.

4.2 Hypothesis testing

A hypothesis test asks: **is what I see just chance, or is there a real effect?** You set up two competing claims.

- **Null hypothesis H_0**: no effect, no difference (the default assumption).
- **Alternative hypothesis H_1**: there is an effect.

The recipe:

- Choose a significance level **α** (commonly 0.05) in advance.
- Compute a **test statistic** from the data, such as a z or t value.
- Find the **p-value**: the probability, *assuming H_0 is true*, of getting a result at least as extreme as yours.
- If $p < \alpha$, **reject** H_0. Otherwise **fail to reject** it.

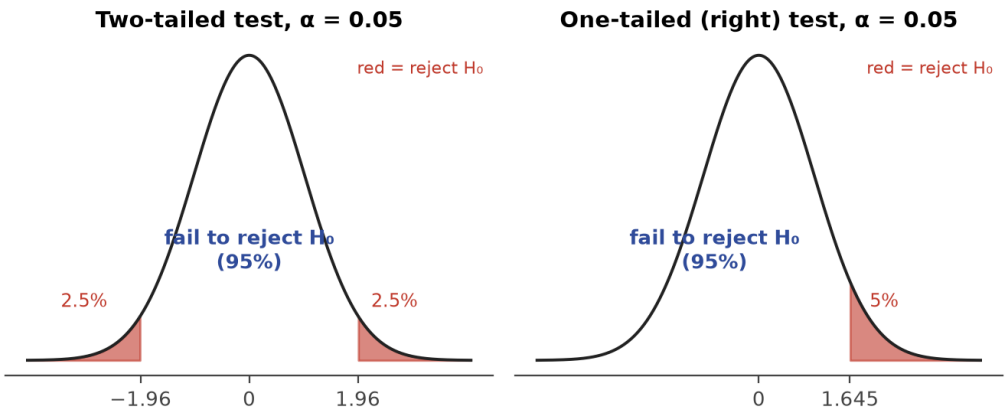


Figure 4.2 · Red areas are the rejection regions. A statistic that lands there is too unlikely under H₀. Left: two-tailed. Right: one-tailed.

WATCH OUT

What a p-value is not. p = 0.03 does not mean "a 3% chance H_0 is true". It means "if H_0 were true, data this extreme would appear 3% of the time". And failing to reject is not proof that H_0 is true; it only means the evidence was not strong enough.

	H_0 is actually true	H_0 is actually false
Reject H_0	**Type I error** (false positive), probability α	Correct. Probability = **power** = 1 − β
Fail to reject H_0	Correct	**Type II error** (false negative), probability β

One-tailed vs two-tailed. A two-tailed test asks "is it different?" (either direction) and splits α between the tails. A one-tailed test asks "is it larger?" (or smaller) and puts all of α in one tail. Decide before you look at the data. **Power** rises with a larger sample, a larger true effect, or a larger α.

IN YOUR LOAN-DEFAULT PROJECT

Missing vs present. The gap of about 1.5 percentage points in default rate for EXT_SOURCE_3 comes with a standard error of about 0.12 points: more than 10 standard errors, so the p-value is essentially 0 and the difference is clearly not chance.

IN YOUR LOAN-DEFAULT PROJECT

Many tests at once. If you test 120 columns at α = 0.05 and none has a real effect, the chance of at least one false positive is $1 - 0.95^{120} \approx$ **99.8%**, and you would expect about 6 by luck. A simple fix (Bonferroni) tests each at α / 120 ≈ 0.0004. This is why a borderline "significant" result on a tiny group should not be trusted.

4.3 The t distribution and t-tests

When σ is unknown we estimate it with s. The statistic then follows a **t distribution**, which looks like the normal but with heavier tails, reflecting the extra uncertainty. It is controlled by the **degrees of freedom** (n − 1 for one sample) and approaches the normal as the degrees of freedom grow (about 30 and above).

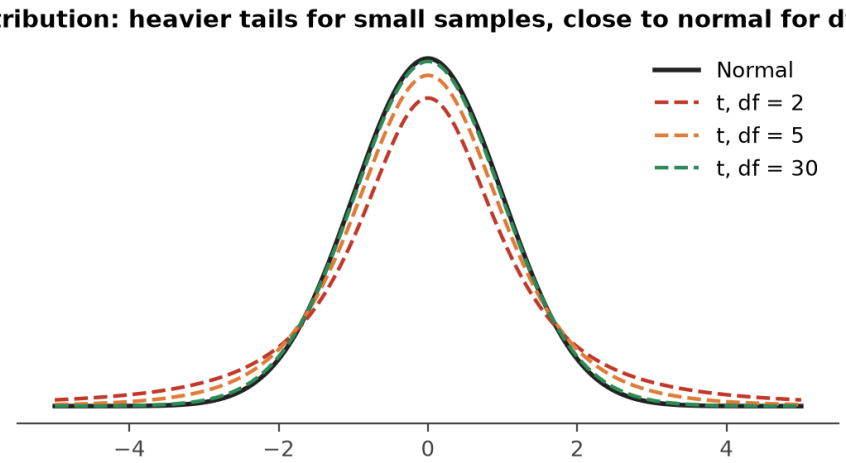


Figure 4.3 · With few degrees of freedom the t curve has fat tails; by 30 it is almost the normal curve.

Test	Use it when…	Statistic
One-sample t-test	Comparing a sample mean with a claimed value μ_0	$t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $df = n - 1$
Two-sample t-test	Comparing the means of two *independent* groups	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
Paired t-test	Before/after on the *same* subjects, or matched pairs	A one-sample test on the differences

EXAMPLE

One-sample. A process should average $\mu_0 = 50$. A sample of $n = 16$ has $\bar{x} = 52$ and $s = 12$. SE = 3, so $t = (52 - 50)/3 = 0.67$ with 15 degrees of freedom. The two-tailed p-value is about 0.51, far above 0.05: no evidence the mean differs from 50.

KEY IDEA

Paired vs two-sample. If the same people are measured twice, use the paired test. It removes person-to-person variation and is usually more powerful. Using a two-sample test on paired data throws that advantage away.

4.4 A/B testing

An **A/B test** compares two versions (a web page, a model, a treatment) by randomly assigning users and comparing a metric, often a conversion *proportion*. It is a two-sample hypothesis test for proportions:

$$z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}\,(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$$

where $\hat{p}$ is the pooled proportion across both groups.

EXAMPLE

Worked example. Version A: 200 conversions out of 5,000 (4.0%). Version B: 240 out of 5,000 (4.8%).
Pooled $\hat{p} = 440/10{,}000 = 0.044$. SE $= \sqrt{(0.044 \times 0.956 \times (1/5000 + 1/5000))} \approx 0.0041$.
$z = (0.048 - 0.040)/0.0041 \approx$ **1.95**, and the two-tailed p-value ≈ **0.051**.
At $\alpha = 0.05$ we **fail to reject** H_0, though only just. B looks better, but the evidence is borderline; collecting more data would settle it.

WATCH OUT

Pitfalls. Decide the sample size and the metric *before* the test. Stopping early the moment p dips below 0.05 (“peeking”) inflates false positives. A result can be statistically significant yet too small to matter in practice, so report the size of the effect with its confidence interval, not only the p-value.

IN YOUR LOAN-DEFAULT PROJECT

Your missing-vs-present comparisons used exactly this structure: two groups, a gap in rates, a standard error, and a rule for deciding whether the gap is real. That is a two-proportion test.

Check yourself (answers in Appendix C)

- How should a 95% confidence interval be interpreted correctly?
- What does p = 0.03 mean, and what does it not mean?
- What are Type I and Type II errors?
- Why use the t distribution instead of the normal when σ is unknown?
- In the A/B example (z = 1.95, p = 0.051), what is your decision at α = 0.05?

Appendix A · Formula cheat sheet

Topic	Formula
Complement	$P(\text{not } A) = 1 - P(A)$
Sum rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
Independence	$P(A \text{ and } B) = P(A)\,P(B)$
Conditional probability	$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$
Bayes' theorem	$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$
Binomial	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
Binomial mean / variance	$E[X] = np, \qquad \mathrm{Var}(X) = np(1-p)$
Expected value	$E[X] = \sum_x x\,p(x) \qquad E[X] = \int x\,f(x)\,dx$
Linearity	$E[aX + b] = aE[X] + b, \qquad E[X + Y] = E[X] + E[Y]$
Variance	$\mathrm{Var}(X) = E\left[(X-\mu)^2\right] = E[X^2] - \mu^2$
z-score	$z = \frac{x - \mu}{\sigma}$
Covariance	$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$
Correlation	$\rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$
Sample mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Sample variance	$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$
CLT and standard error	$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right), \qquad SE = \frac{\sigma}{\sqrt{n}}$
Likelihood	$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$
MAP estimate	$\hat{\theta}_{MAP} = \arg\max_\theta \; [\log p(x \mid \theta) + \log p(\theta)]$
Ridge loss	$\min_w \sum_i (y_i - w^T x_i)^2 \; + \; \lambda \lVert w \rVert^2$
CI for a mean	$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$
CI with t	$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$
CI for a proportion	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Sample size	$n = \left(\frac{z^* \sigma}{ME}\right)^2$
One-sample t	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Two-proportion z	$z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}\,(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$

Appendix B · Glossary

Term	Meaning
Sample space	The set of all possible outcomes.
Independent events	One does not change the probability of the other: P(A and B) = P(A)P(B).
Prior / likelihood / posterior	Belief before data / how well parameters explain the data / belief after data.
Random variable	A rule that assigns a number to each outcome.
PMF / PDF / CDF	Probability of a value (discrete) / density (continuous) / cumulative probability up to x.
Expected value	The probability-weighted average of a random variable.
Variance, standard deviation	Average squared deviation from the mean; its square root.
Skewness, kurtosis	Asymmetry of a distribution; heaviness of its tails.
Quantile, IQR	Value below which a share of the data lies; the span of the middle half.
Marginal / conditional	Distribution of one variable alone / of one given the other.
Covariance, correlation	How two variables move together; the unit-free version in $[-1, 1]$.
Population, sample	Everything of interest; the observed part.
Parameter, statistic	A population quantity; a number computed from a sample.
Standard error	Standard deviation of an estimate, such as $\sigma/\sqrt{n}$ for a mean.
CLT	Sample means are approximately normal for large n.
MLE / MAP	Parameter that maximises the likelihood / the posterior.
Regularization	A penalty on weights that guards against overfitting.
Confidence interval	A range built to capture the parameter in a stated share of repeated samples.
p-value	Probability, if H_0 is true, of a result at least as extreme as observed.
Type I / Type II error	Rejecting a true H_0 / failing to reject a false one.
Power	Probability of rejecting H_0 when it is false $(1 - \beta)$.
A/B test	A randomised comparison of two versions, usually tested with a two-sample test.

Appendix C · Answers to “Check yourself”

Week 1

1. The complement needs one product, $(5/6)^4 \approx 0.482$, instead of counting every way to get one or more sixes. Answer: $1 - 0.482 = 0.518$.
2. $P(A \mid B) = P(A)$.
3. Prior $P(\text{spam}) = 0.30$; likelihood $P(\text{free} \mid \text{spam}) = 0.60$; posterior $P(\text{spam} \mid \text{free}) \approx 0.837$.
4. A single point has zero width, so it has zero area under the density. A range has width, so it has area, which is its probability.
5. Mean $np = 3$; variance $np(1 - p) = 10 \times 0.3 \times 0.7 = 2.1$.

Week 2

1. $E = 3.5$, variance $= 35/12 \approx 2.92$.
2. The mean is larger, because it is pulled toward the long right tail.
3. $\text{Cov} = 0$. The converse is false: $y = x^2$ has covariance (and correlation) near 0 but is fully dependent.
4. Correlation is covariance divided by both standard deviations, so it has no units and lies between -1 and 1.
5. $0.0185 / 0.198 \approx 9.3\%$.

Week 3

1. $\bar{x}$ is estimated from the same data, which makes deviations from it a little too small. Dividing by $n - 1$ removes that bias.
2. It is approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$.
3. $7/10 = 0.7$.
4. MLE plus an L2 (ridge) penalty.
5. It halves, since $SE = \sigma/\sqrt{n}$ and $\sqrt{4} = 2$.

Week 4

1. If you repeated the sampling many times, about 95% of such intervals would contain the true value. It is a statement about the procedure.
2. If H_0 were true, data this extreme would occur 3% of the time. It is not the probability that H_0 is true.
3. Type I: rejecting a true H_0 (false positive). Type II: failing to reject a false H_0 (false negative).
4. Because s only estimates σ, adding uncertainty. The t distribution has heavier tails to account for it.
5. Fail to reject H_0 ($p \approx 0.051$ is just above 0.05). The evidence is borderline, so collect more data.

Appendix D · Where each idea appeared in the loan-default project

Concept	Where it showed up
Conditional probability	Default rate when a column is missing vs present: P(default | missing).
Independence	Is a missing flag independent of default? The gap says no.
Bernoulli / binomial	Default is Bernoulli(≈ 0.081); defaults in a group are binomial.
Standard error and the normal approximation	se_pts and the “gap > 2 × SE” rule.
Skewness, log transform	AMT_INCOME_TOTAL and other money columns.
Quantiles, outliers	p99 and max-over-second screen for extreme values.
Covariance and correlation	Spearman of bureau counts vs EXT_SOURCE_3 (−0.019).
Confidence interval for a proportion	Why a 13-row minority cannot be trusted.
Hypothesis test, multiple testing	Screening about 120 columns; some “real” gaps arise by chance.
MLE / regularization	Logistic regression and ridge/lasso in the modelling phase.